

How to use the United Kingdom tool

A plain-language guide
BENEFITS D2.2 Microsimulation model

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1 What this tool is, and what it is not

This is the United Kingdom tool from the BENEFITS project. It is a single web page. You open the file in an ordinary web browser; there is nothing to install, you do not need to be online, and there is no login. There are four such tools, one each for the United Kingdom, Spain, Italy, and Greece. They all work the same way and differ only in the country they describe, so once you know one you know all four.

Two things are worth knowing before you start. First, the tool does not run a live tax and benefit model while you use it. All the numbers were worked out in advance, checked, and saved inside the page. When you move a control, the tool looks up the saved results and fills in the points in between. This is why it is fast and why it holds no personal data. It also means the tool can only show what was worked out beforehand: it fills in between the marked points on a slider, and it stops at the ends of each slider rather than guessing past them.

Second, the values on the Provider tab are not measurements of your own clients. They take findings from published research and apply them to the caseload you describe. Every value comes with a low, a middle, and a high figure, and you should always quote the range, not just the middle number, because the middle number on its own claims more certainty than the evidence supports.

2 Opening it and finding your way around

Open the tool's web page in a browser such as Chrome, Edge, Firefox, or Safari. It fills the window. Nothing you do is saved, so you can explore freely and reload the page to start over. Across the top are four tabs:

- **Home** tells you what the tool is, where its data comes from, and the warnings to read before you quote anything.
- **Analyst** lets you compare a reform or a shock: who gains or loses, the effect on government money, and the effect on poverty.
- **Provider** lets you put a value on a service: describe a caseload and read its social value, its value for money, and how a downturn would change demand.

- **Scenarios** is a fixed menu of ready-made reforms and shocks for the United Kingdom, each one a saved result you can read straight off.

3 If you read only one thing

These are planning estimates, not budgets. Some figures are known to be overstated, and the tool says which. The capacity and downturn figures should not be used to set staffing: they carry the same assumptions in every tool and none of them is calibrated to your area. Most of the service costs are assumptions, and the tool marks which ones with stars. Where you have your own figures, type them into the Provider tab and the tool will use yours instead of ours.

4 The Analyst tab: comparing reforms and shocks

Move one control at a time. Each slider is one reform or one shock. When you move a slider, the others jump back to their starting point. This is deliberate: two reforms happening together do not simply add up, and the tool will not pretend they do, so it shows you one at a time. The small ticks on each slider mark the points that were actually worked out; between the ticks the tool draws a straight line, so you can always see where the real results sit and where the tool is filling in.

This tool has seven sliders and one on/off toggle. Here is what each one does:

- **Minimum income (Universal Credit standard allowance):** raises the basic monthly Universal Credit payment. The slider models increases only, so it does not run below the current rate.
- **Child or family element (Child Benefit):** raises Child Benefit.
- **Personal allowance (income tax threshold):** changes how much you can earn before you start paying income tax.
- **GDP shock (market income):** an economic downturn, which this tool passes through to unemployment. The three European tools have an “Earnings downturn” slider instead, which is a different experiment.
- **Unemployment shock:** a direct rise in the unemployment rate.
- **UC taper rate:** how fast Universal Credit is withdrawn as earnings rise.
- **UC work allowance:** how much someone can earn before that withdrawal starts.
- **Toggle: remove the High Income Child Benefit Charge:** switches off the charge that takes Child Benefit back from higher earners. Like the sliders, turning it on resets them.

Set the fairness weight. There is one slider, labelled with the Greek letter ε (epsilon), that asks you to make a judgement. It decides how much extra weight to give to gains that reach lower-income households. At the lowest setting, 0, a £1 counts the same wherever it lands, and the headline is a simple total of gains and losses. As you raise it to 1 and then 2, gains to poorer households count for more. Because two reforms can swap places as you raise this slider, the tool draws each result as a line across the range rather than a single number. Where two lines cross, the better reform changes. Quote the crossing point, not one spot on the line.

Read the three results, which are kept separate on purpose.

- **Who gains or loses.** The headline is the overall gain or loss in £, after the fairness weight is applied. A bar chart splits it across ten income groups from poorest to richest, and the bars

add up to the headline, so you can see which groups drive it. A simple winners-and-losers count sits beside it.

- **Government money.** Separate lines show the tax and benefit money flowing in and out.
- **Poverty.** Two counts. The main one uses a poverty line fixed at the starting point and held still, so it counts people crossing a standard that does not move. A second one uses a line that moves with average income. The two can disagree about whether a tax rise raises or lowers poverty, and the tool explains why on screen. Lead with the fixed-line count.

One caution the tool repeats, and you should carry it: reforms are shown without anything to pay for them, so nobody is shown losing from the cost. Treat any figure as a first-round effect, not a fully funded package.

5 The Provider tab: putting a value on a service

Describe the caseload. Pick a service, enter how many people it helps in a year, choose roughly where those people sit across the income range, and, if you want a value-for-money figure, enter the running cost. You can also enter the size of a downturn. The fairness-weight slider works here too. Everything the tool reports back is counted per person helped, so enter numbers of people rather than households or families. The running cost is the exception: enter the total cost of running the service over the year, not a cost per case.

Use your own cost if you have one. Each service arrives with a cost per person already filled in. Those are our figures and not yours, and for most services they are informed estimates rather than measured costs, which is the honest position: we searched for published service costs across Europe and found very few. If you know what a case actually costs you in the United Kingdom, type it in and the tool uses yours from that point on. You do not have to: leave the box alone and the tool works exactly as it does now. A figure that rests on a cost you entered is marked with a dagger (†) rather than our stars, it stays marked in anything you export or share, and it is your figure rather than ours, so do not quote it as a result from the model.

Read the ladder from the bottom up.

- The **cash floor** is the cautious lowest figure: the income effect only, weighted for who receives it.
- The **headline value** adds the good the service does, drawn from published findings. It is not the full value of the outcome for everyone you serve. It applies the share of a caseload that published evidence says reaches the outcome, and then the share of that outcome the evidence attributes to the service rather than to everything else in someone's life. Money-type outcomes are weighted for who benefits; wellbeing outcomes are counted at a standard value for a year of improved wellbeing (the tool calls this the WELLBY value).
- **Enter the caseload the evidence measures.** Some of those shares are measured on a narrower group than everyone a service sees. Where that is so, the tool names the group on the service's own note, and entering a wider caseload will overstate the result.
- The **low, middle, and high range** is genuine uncertainty from the evidence. Carry it into any funding case.
- The **value-for-money ratio** is the middle headline divided by the running cost you entered: social value for each £1 spent.

The downturn view projects the extra demand a recession would place on your service, scaled to your caseload, and shows the staff and budget needed to meet it. Read it as rough planning, not a forecast for your area, for two reasons the tool states: the numbers behind it are informed estimates and not nationally measured figures; and above a downturn of five percentage points the view draws a straight line up to ten, marking every figure in that stretch as outside the range that was actually modelled.

6 The star marks

Some figures carry star marks, and the key sits on every panel so it stays with a screenshot. No star means the figure rests only on United Kingdom sources, which after the evidence review is rare: most values in this tool carry a mark. One star (*) means it uses something borrowed from another country and adjusted to fit. Two stars (**) mean it uses something borrowed without adjustment, or an assumption. A figure takes the worst mark of the things it is built from. One point is easy to misread: on the downturn view every figure carries two stars, because the staff numbers and unit costs behind it are assumptions. A tool with fewer stars is not more reliable; the stars show where a number was borrowed, not where it is well proven. There is also a dagger (†), which is not one of ours: it marks a figure that rests on a unit cost you typed in yourself, and it appears instead of our stars because the reason for those stars no longer applies to your own number.

7 Things to keep in mind for United Kingdom

The Home tab carries the full list; these are the ones that most affect how you read a figure.

- This tool counts working-age benefit units only, which it labels “benefit units” on screen, and which in this build are the same thing as households; the three European tools count all households. So figures per unit, value-for-money ratios, and poverty counts are correct within this tool but should not be compared directly with the European tools.
- Two poverty lines are used here. The one the tool leads with is fixed at 60% of the middle (median) income, adjusted for household size, and then held still, which is the same construction the three European tools use. The second, which moves with average income, is the official UK statistical definition, and is built differently from the European tools’ moving line. Either way the counts are not comparable across tools, because this tool counts people in working-age benefit units and the European tools count everyone.
- Every reform can be viewed with the High Income Child Benefit Charge on or off, using the toggle.
- Downturn figures from this tool are built differently from those in the other BENEFITS tools and should not be read across them.

8 The Scenarios tab

This tab holds a fixed menu for the United Kingdom: the baseline plus fifteen ready-made reforms and shocks. Each is a saved result rather than an in-between estimate, so you can read it straight off. They include the paired comparisons discussed in the technical report, which for this tool set a shock against the response to it. Read the line across the fairness-weight range rather than a single point: which of the two comes out ahead changes as the weight rises, and for the United Kingdom the two lines cross at a fairness weight of about 1.62. That crossing point is the finding, not any single reading off either line.

The one thing to remember: the range and the star marks travel with every figure. The cash floor is a floor, the headline is a weighted estimate built partly on borrowed evidence, and the fairness weight is a judgement you set, not a result the tool dictates.

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